CS & IT ENGINEERING

DIGITAL LOGIC

MINIMIZATION



Lecture No. 5



By - CHANDAN SIR

Recap of Previous Lecture



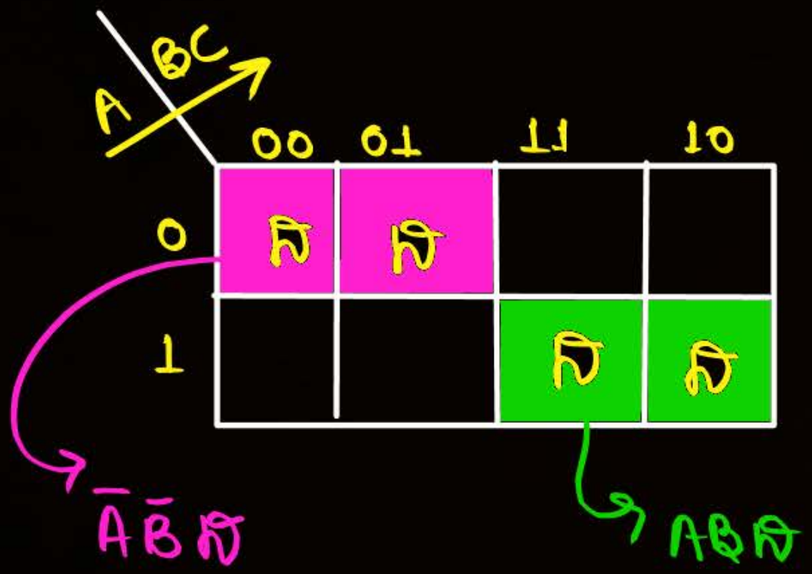
K-MAP

Topics to be Covered



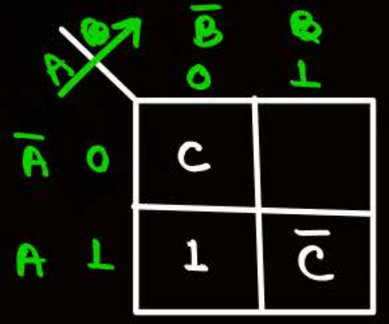
PRIME IMPLICANTS AND ESSNTIAL PRIME IMPLICANTS

- PRIME IMPLICANTS
- ESSENTIAL PRIME IMPLICANTS
- Questions Practice
- Discussion



$\bar{A}\bar{B}\bar{C} + AB\bar{C}$

Q.

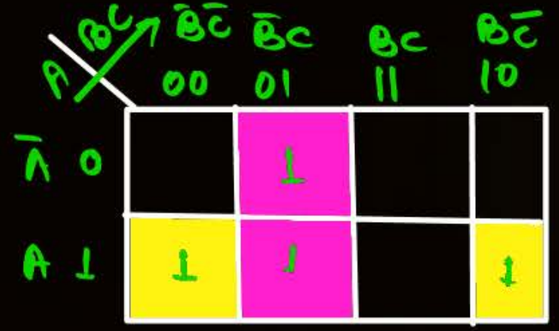


$f = \bar{A}\bar{B}C + A\bar{B} + AB\bar{C}$

$f = \bar{A}\bar{B}C + A\bar{B}(\bar{C}+C) + AB\bar{C}$

$= \bar{A}\bar{B}C + A\bar{B}\bar{C} + A\bar{B}C + AB\bar{C}$

$= \Sigma m(1,4,5,6)$



$= A\bar{C} + \bar{B}C$ Ans

By Boolean algebra

$\bar{A}\bar{B}C + A\bar{B} + AB\bar{C}$

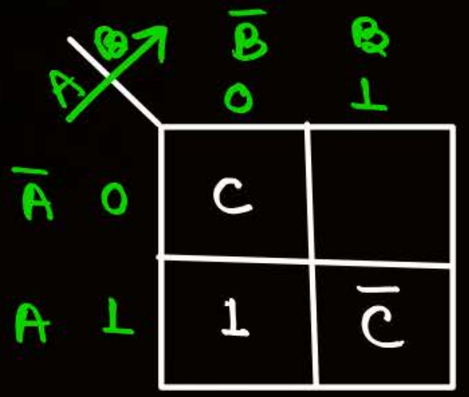
$\bar{A}\bar{B}C + A\bar{B} + A\bar{B} + AB\bar{C}$

$\bar{B}[\bar{A}C + A] + A[\bar{B} + B\bar{C}]$

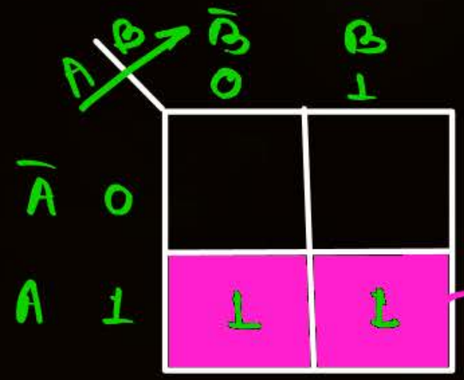
$\bar{B}[A+C] + A[\bar{B}+\bar{C}] = A\bar{B} + \bar{B}C + A\bar{B} + A\bar{C} = A\bar{B} + \bar{B}C + A\bar{C}$

$= \bar{B}C + A\bar{C}$

Q.

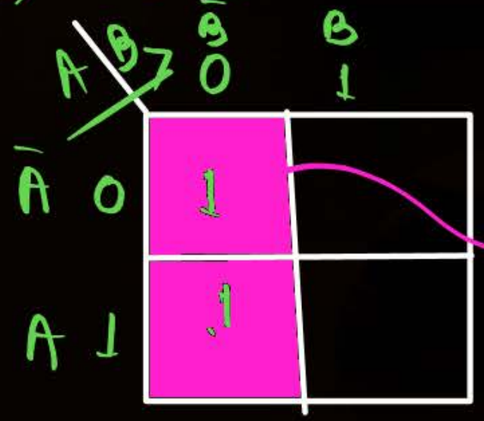


Case (1) $C = 0$ ✓



$\Rightarrow A\bar{C}$

Case (2) $C = 1$

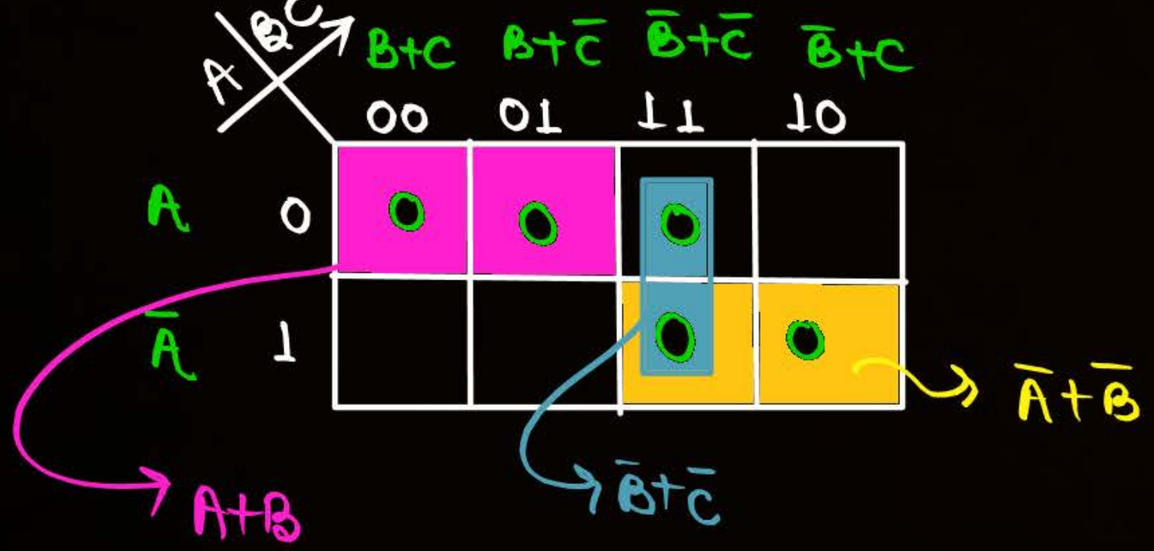


$\bar{B}C$

$\Rightarrow A\bar{C} + \bar{B}C$

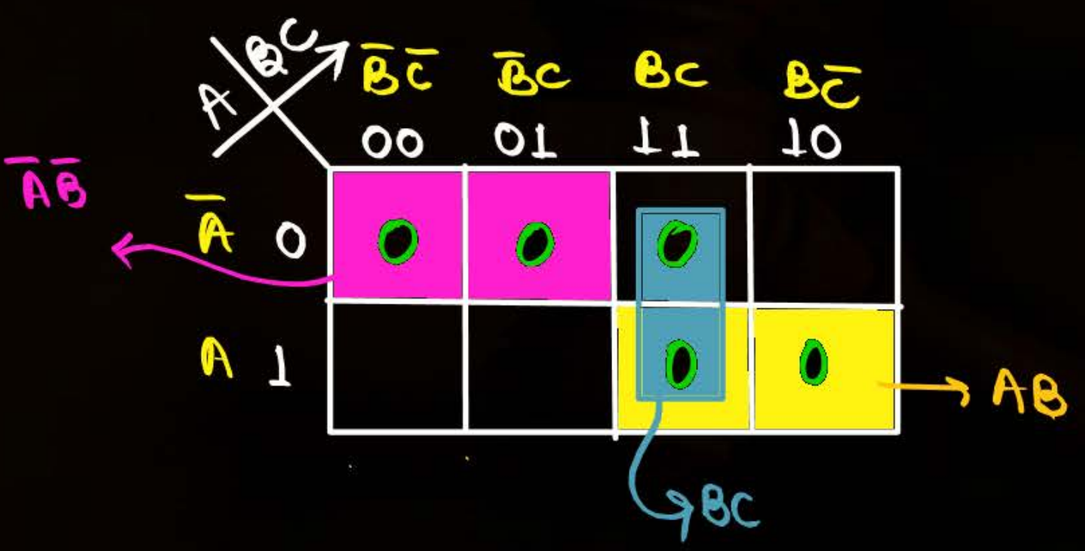
Q $F(A,B,C) = \pi M(0,1,3,6,7)$

Method 1.



$(A+B)\cdot(\bar{B}+\bar{C})\cdot(\bar{A}+\bar{B})$

Method 2.



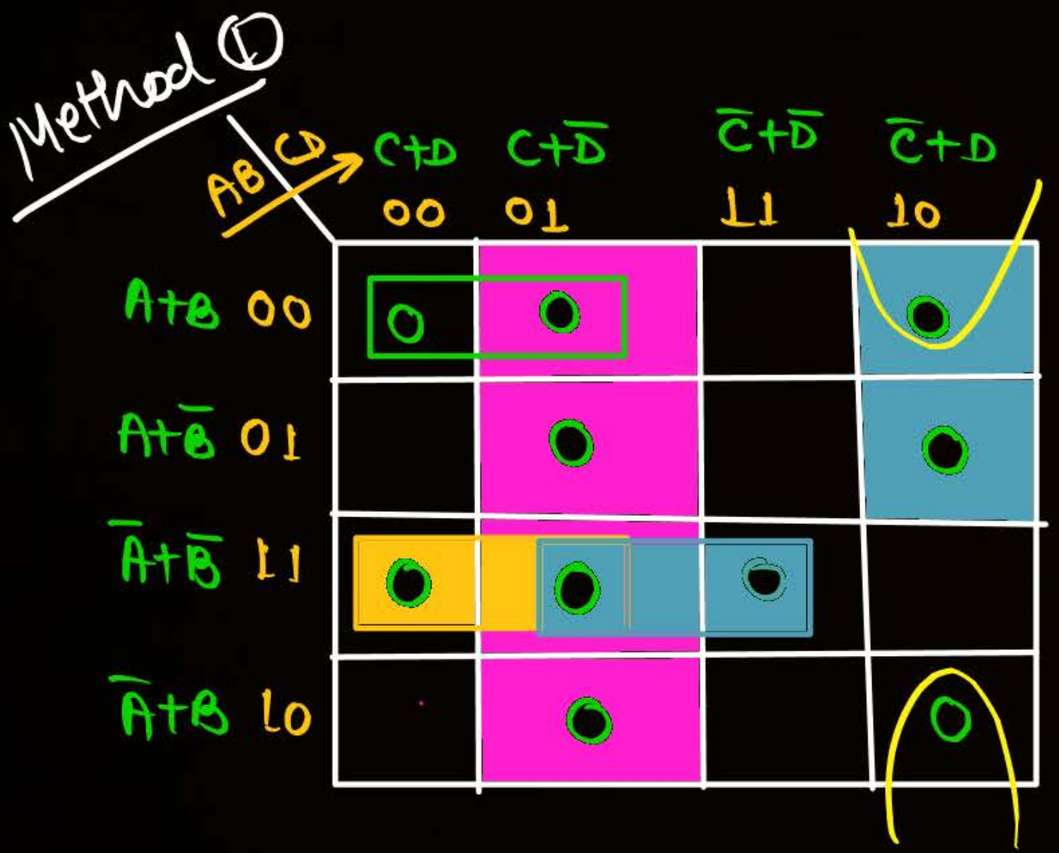
$\bar{F} = \bar{A}\bar{B} + Bc + AB$

$F = \overline{\bar{A}\bar{B} + Bc + AB}$

$= (A+B)\cdot(\bar{B}+\bar{C})\cdot(\bar{A}+\bar{B})$

Q $F = \Pi M(0, 1, 2, 5, 6, 9, 10, 12, 13, 15)$

Method ①

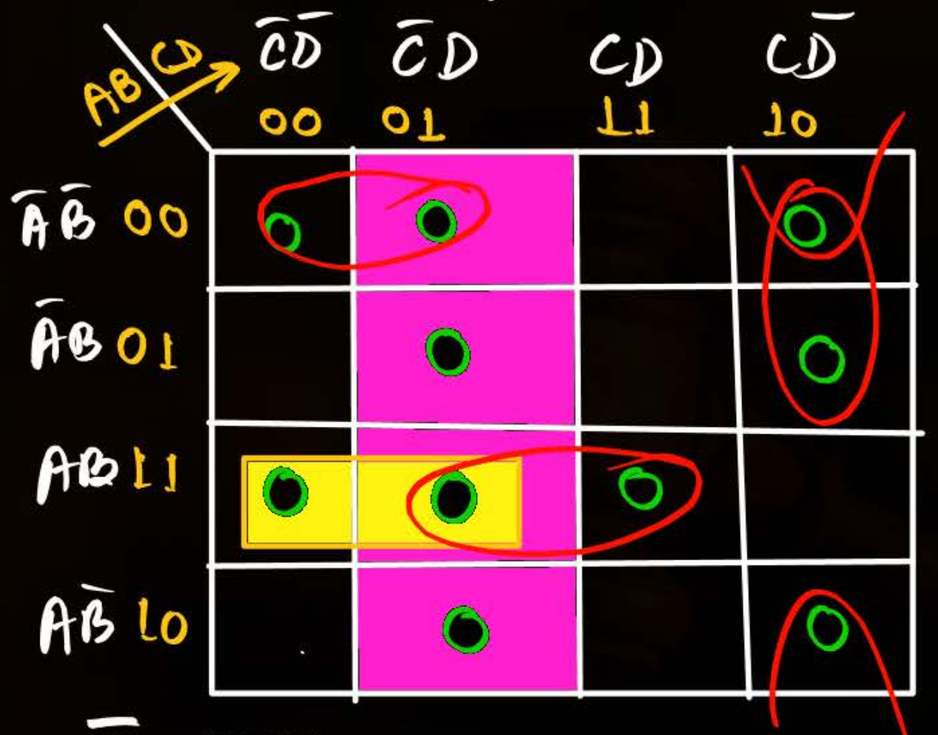


$F = (A+B+C)\cdot(C+\bar{D})\cdot(\bar{A}+\bar{B}+C)\cdot(\bar{A}+\bar{B}+\bar{D})\cdot(A+\bar{C}+D)\cdot(B+\bar{C}+D)$

$\downarrow$

$(A+B+D)\cdot(\quad)\cdot$

Method ②



$\bar{F} = \bar{A}\bar{B}\bar{C} + \bar{C}D + AB\bar{C} + ABD + \bar{A}C\bar{D} + \bar{B}C\bar{D}$

$F = (A+B+C)\cdot(C+\bar{D})\cdot(\bar{A}+B+C)\cdot(\bar{A}+\bar{B}+\bar{D})\cdot(A+\bar{C}+D)\cdot(B+\bar{C}+D)$

HW

Q $F(A,B,C,D) = \Pi M(0,1,2,6,7,9,10,11,12,15)$

Q $F(A,B,C,D) = \Pi M(0,2,4,6,8,10,11,12,13) + \Sigma d(3,5,7,15)$

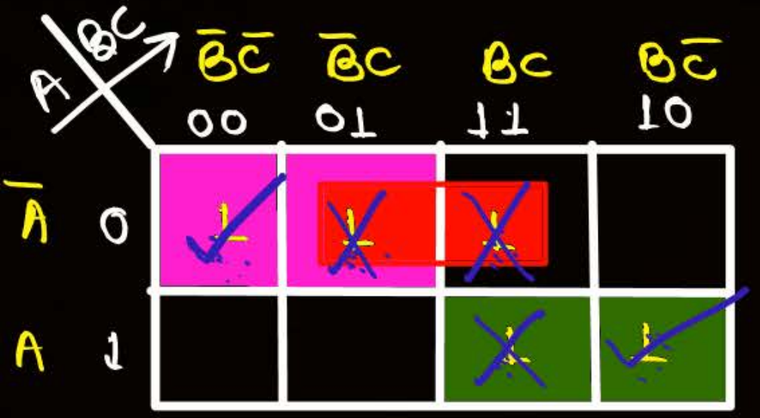


Topic : PRIME IMPLICANTS AND ESSNTIAL PRIME IMPLICANTS

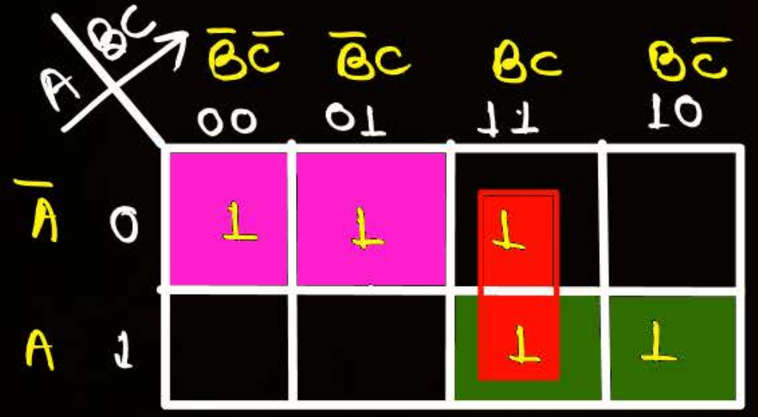
Prime Implicants (PI)

Essential Prime Implicants (EPI)

Q. $f(A,B,C) = \Sigma m(0,1,3,6,7)$



$= \bar{A}\bar{B} + A\bar{B} + \bar{A}C$



$= \bar{A}\bar{B} + A\bar{B} + BC$

$PI = \{\bar{A}\bar{B}, A\bar{B}, \bar{A}C, BC\}$

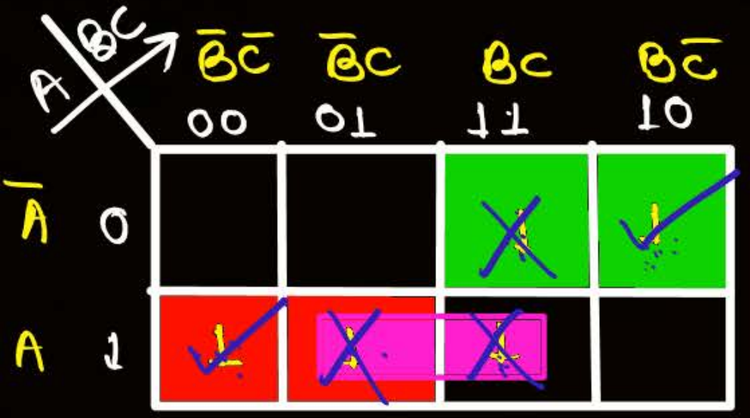
$EPI = \{\bar{A}\bar{B}, A\bar{B}\}$

Implicants = 5

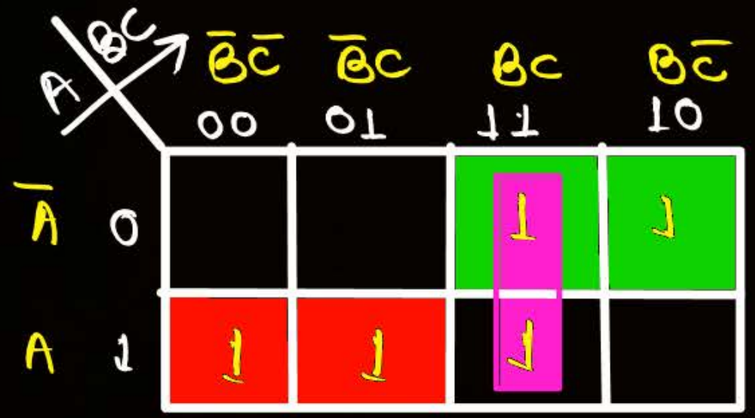
Prime Implicants = 4

Essential Prime Implicants = 2

Ex



$= \bar{A}B + A\bar{B} + AC$ ✓



$= \bar{A}B + A\bar{B} + BC$ ✓

$PI = \{\bar{A}B, A\bar{B}, AC, BC\}$

$EPI = \{\bar{A}B, A\bar{B}\}$

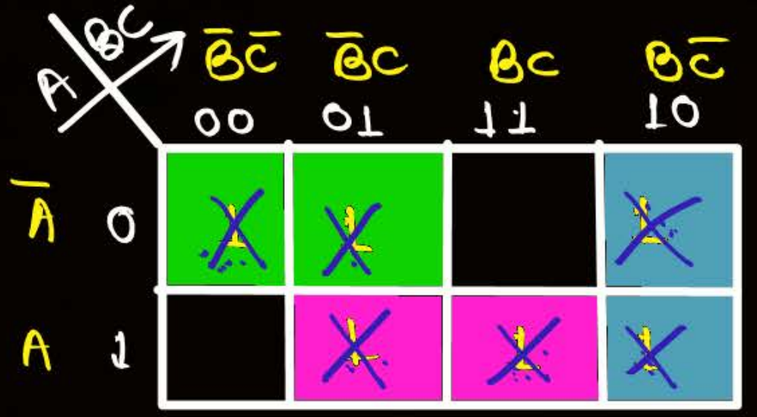
Implicants = 5

PI = 4

EPI = 2

Ex

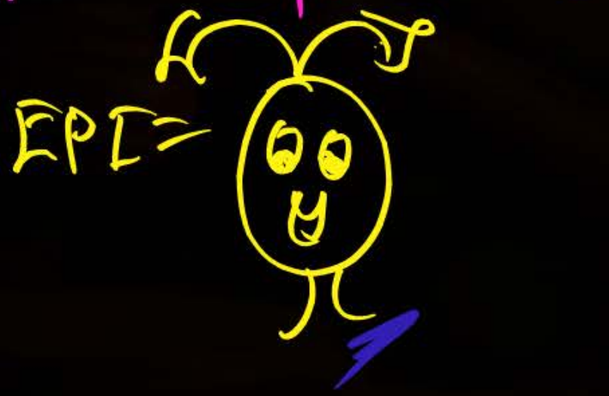
$PI = \{\bar{A}\bar{B}, AC, B\bar{C}, \bar{A}\bar{C}, \bar{B}C, AB\}$

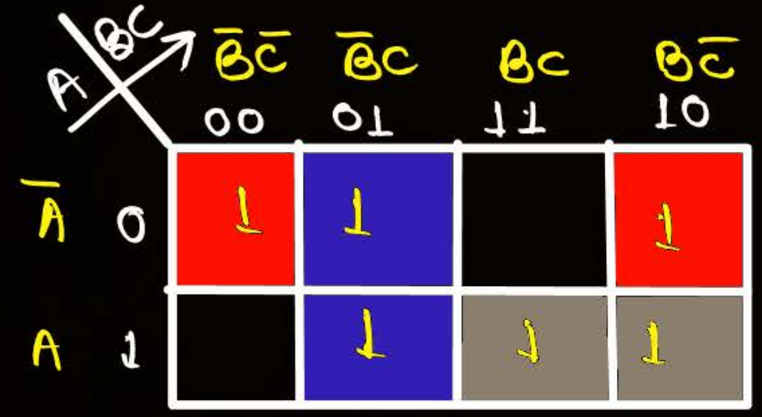


$= \bar{A}\bar{B} + AC + B\bar{C}$

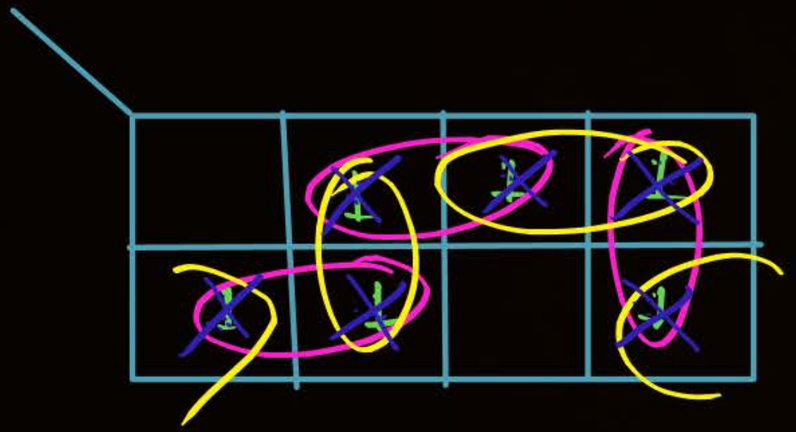
Implicants = 6

prime Implicants = 6





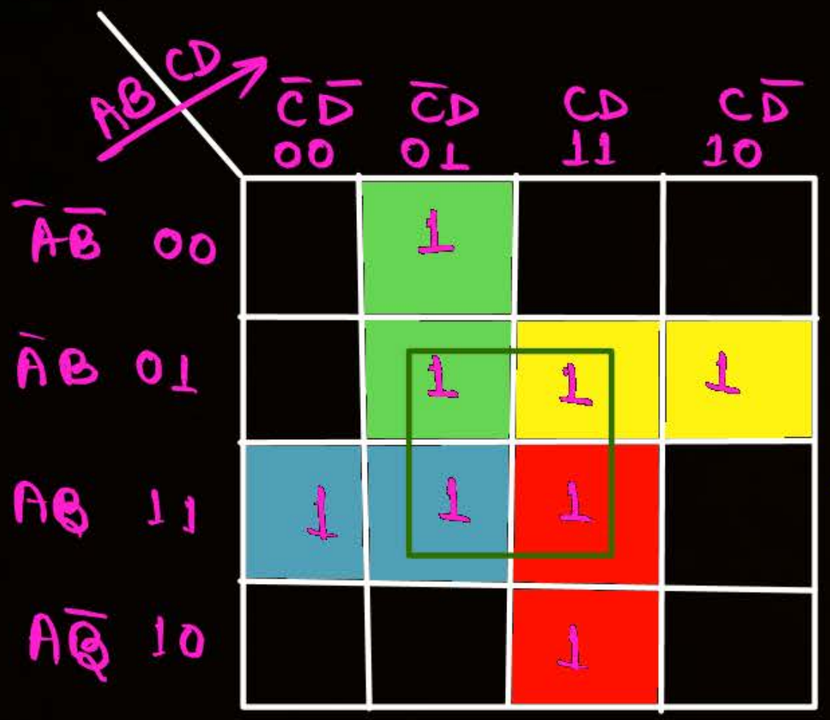
$= \bar{A}\bar{C} + \bar{B}C + AB$



I = 6

PI = 6

EPI = 0



$= \bar{A}\bar{C}D + \bar{A}BC + ACD + AB\bar{C}$ ✓ ✓ ✓ ✓

$PI = \{\bar{A}\bar{C}D, \bar{A}BC, ACD, AB\bar{C}, BD\}$

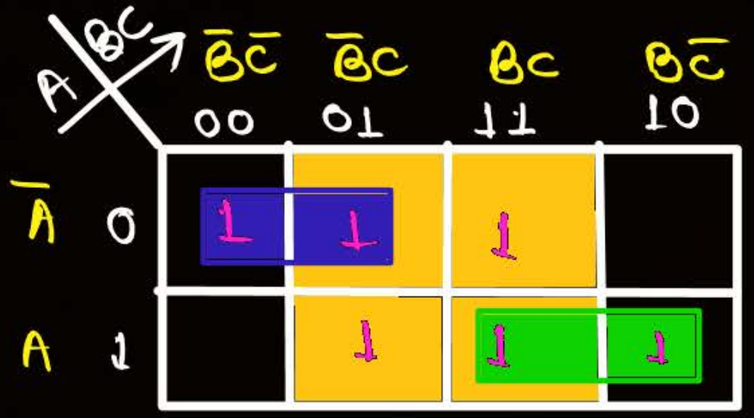
↳ Redendant PI (RPI)

Implicants = 8

prime plicants = 5

EPI = 4

Ex



$= \bar{A}\bar{B} + AB + C$ ✓ ✓ ✓

$PI = \{\bar{A}\bar{B}, AB, C\}$

Implicants = 6

Prime Implicants = 3

EPI = 3



PRIME IMPLICANTS AND ESSNTIAL PRIME IMPLICANTS

(i) Implicants: The total number of minterms in the Boolean expression are called Implicants.or in K-map the total number of 1 is called Implicants.

(ii) Prime Implicants (Pl):



PRIME IMPLICANTS AND ESSNTIAL PRIME IMPLICANTS

(iii) Essential PI/Selective PI:

(iv) Reduced PI:

Note: For an n-variables Boolean function, the maximum, number of prime implicants is 2(n –1). A group of square or rectangle made up of bunch of adjacent minterms which is allowed by definition of K-Map are called prime implicants(PI) i.e. all possible groups formed in K-Map.

#Q. Find the number of Prime (PI) implicants & Essential prime implicants in the given K

(EPI)

PI=6

EPI=6

PQ \ RS	00	01	11	10
00	1			
01		1		1
11	1			
10		1		1

A. 4, 5

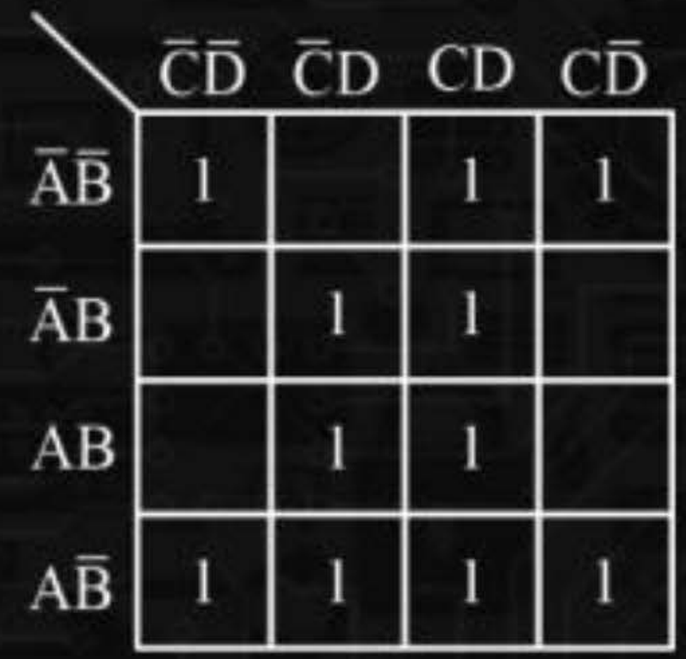
B. 0, 6

C. 6, 0

D. 6, 6

(MCQ)

#Q. Consider the K-map given below:
The EPI term will be:



	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	1		1	1
$\bar{A}B$		1	1	
AB		1	1	
$A\bar{B}$	1	1	1	1

A $BD, A\bar{B}$

B $A\bar{B}, CD$

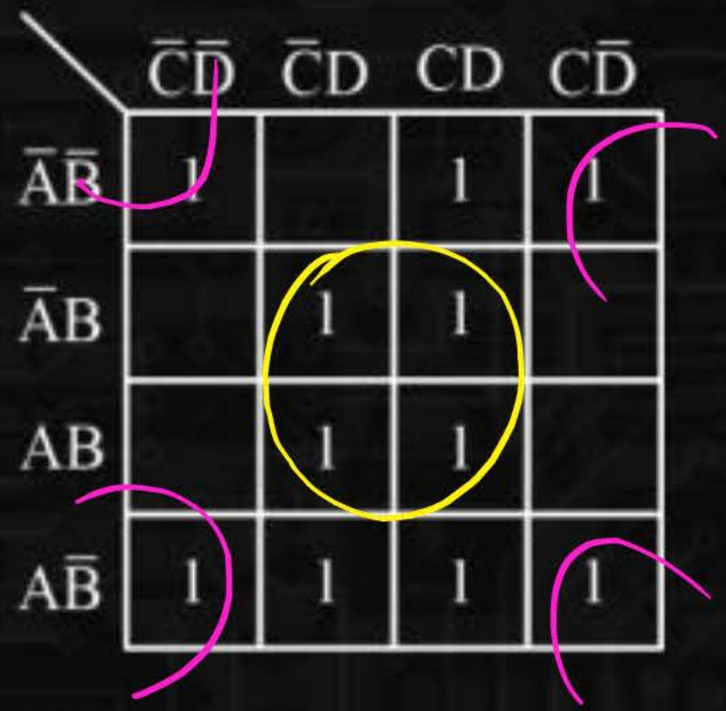
C $BD, \bar{B}\bar{D}$

D $BD, \bar{B}C$

(MCQ)

#Q. Consider the K-map given below:
The EPI term will be:

A	$BD, A\bar{B}$	
B	$A\bar{B}, CD$	
C	$\bar{B}D, \bar{B}\bar{D}$	
D	$BD, \bar{B}C$	

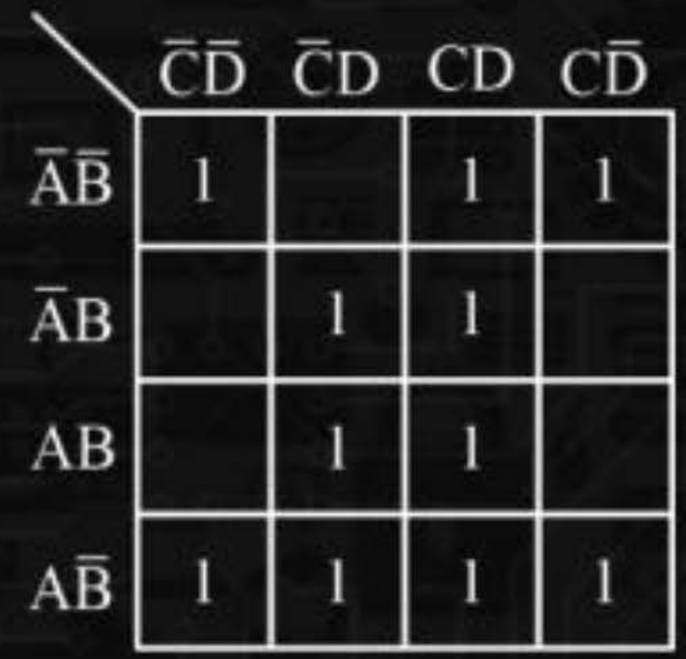


$\bar{B}\bar{D}$, BD

EPI

(MCQ)

#Q. Consider the K-map given below:
The EPI term will be:



	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	1		1	1
$\bar{A}B$		1	1	
AB		1	1	
$A\bar{B}$	1	1	1	1

A $BD, A\bar{B}$

B $A\bar{B}, CD$

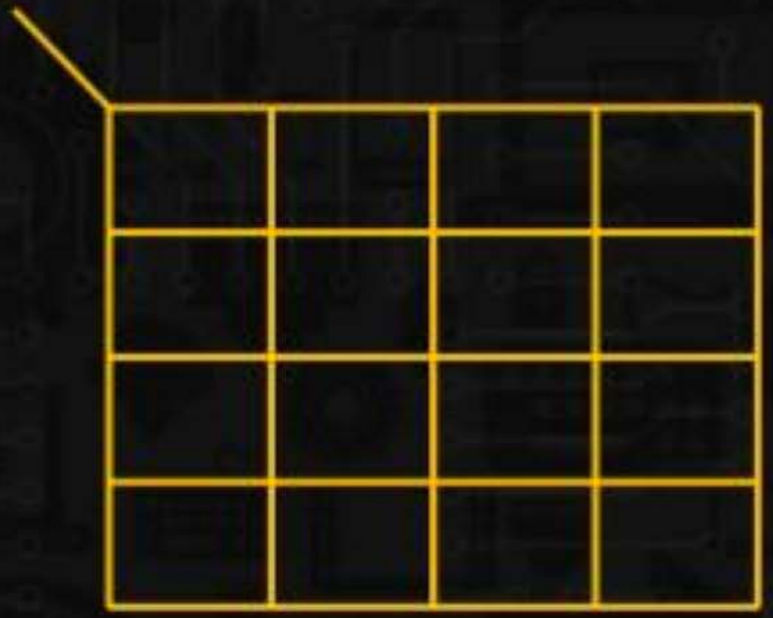
C $BD, \bar{B}\bar{D}$

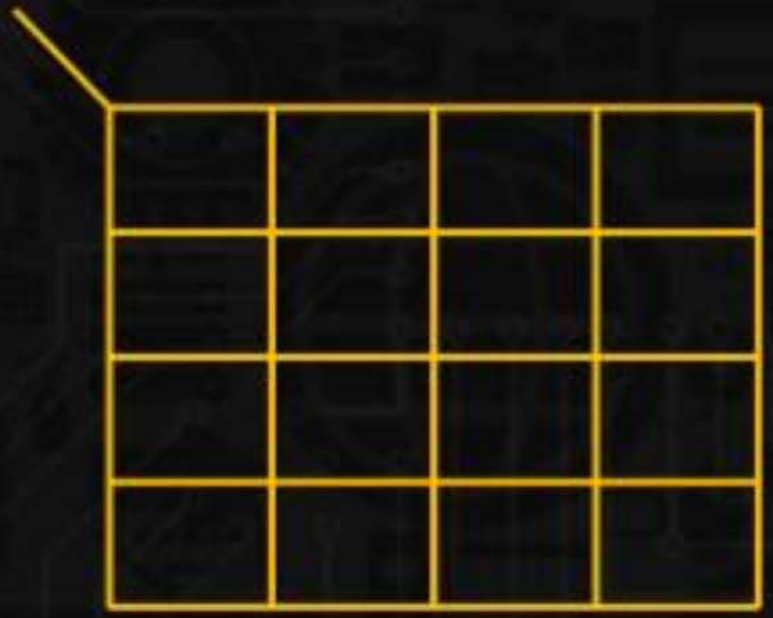
D $BD, \bar{B}C$



5 Variable k-map

#Q. $f(A, B, C, D, E) = \sum m\ (2, 3, 4, 5, 7, 8, 10, 12, 15, 17, 18, 20, 25, 26, 30, 31)$







2 Minute Summary

- PRIME IMPLICANTS
- ESSENTIAL PRIME IMPLICANTS
- Questions Practice
- Discussion

Thank you GW Soldiers !

